5.0 DESIGN PHILOSOPHY, ASSUMPTIONS, AND CONSTRAINTS

The intent for this suite of tools is to develop them on top of a common set of functions and make them available in a Java-based library. These functions are then available to not only the tools but also to the applications and services within the system. The following diagram details where this common library resides in the layered architecture of the system:

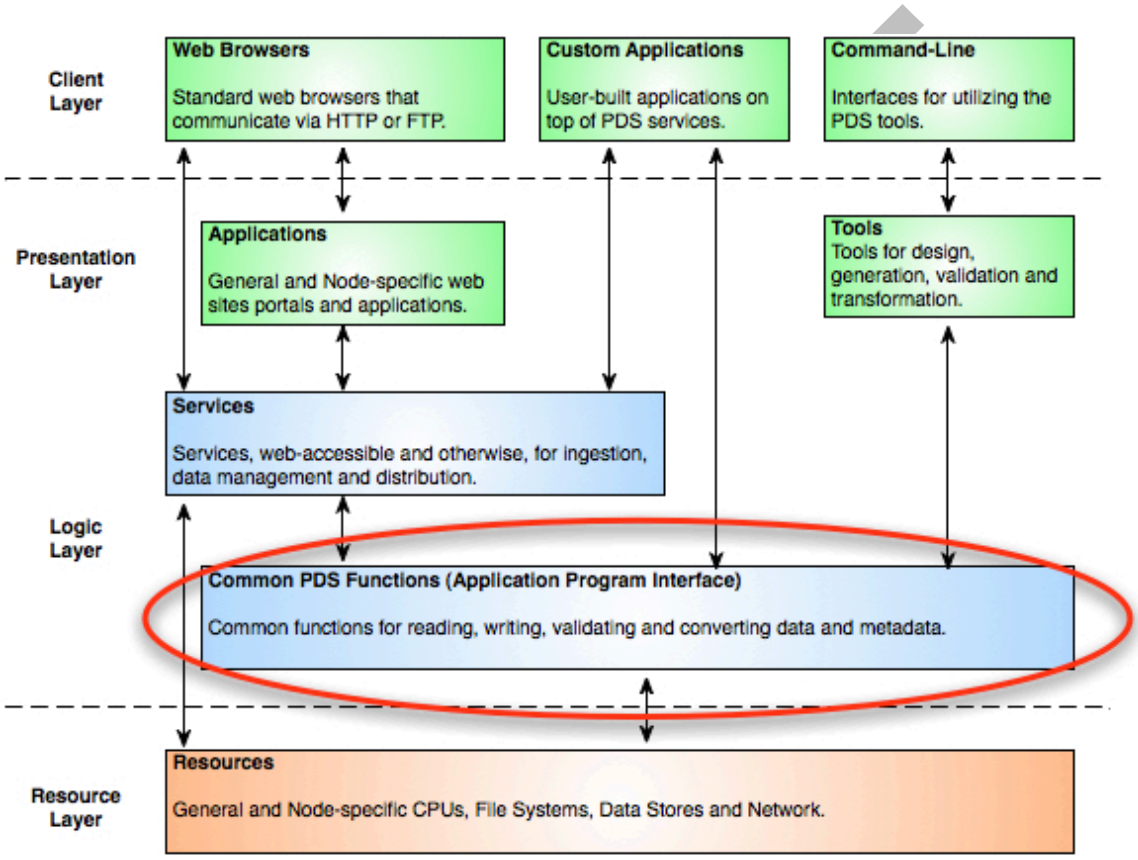

Figure 3: Preparation Tools Approach

Although the majority of this library development will occur at the Engineering Node, portions of the library will be open for contributions from the PDS community. This is especially true when it comes to the transformation function. A few dozen possible transformations were identified in a PDS Node survey [6]. The Engineering Node will look to implement some of the highest priority transformations and invite the PDS community to contribute others.

6.0 ARCHITECTURAL DESIGN

The architectural design covers the component breakdown within the tools, external/internal interfaces and the associated data model.

6.1 Component Architecture

The following diagram details the architecture for the common function library:

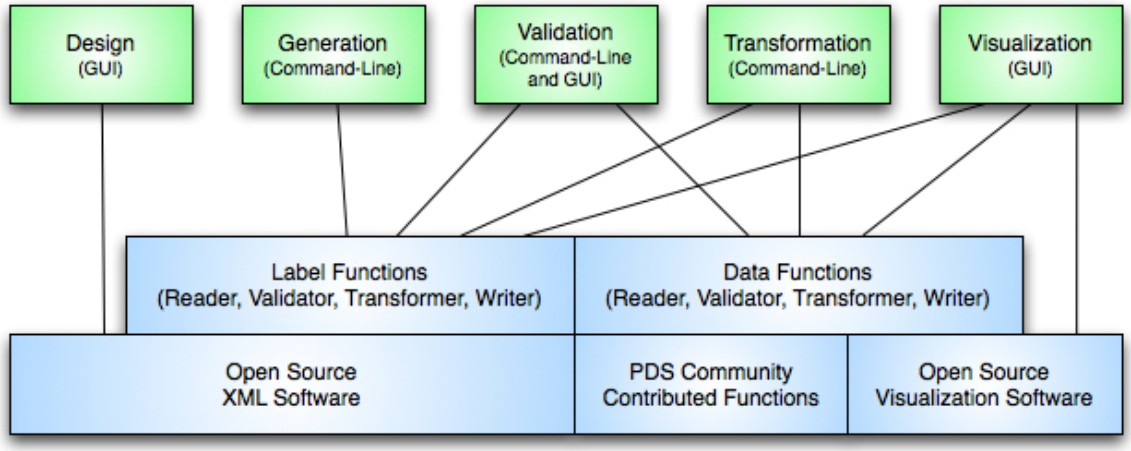

Figure 4: Preparation Tools Library Architecture

6.2 Data Model

The Preparation tools do not have an associated data model.

DRAFT